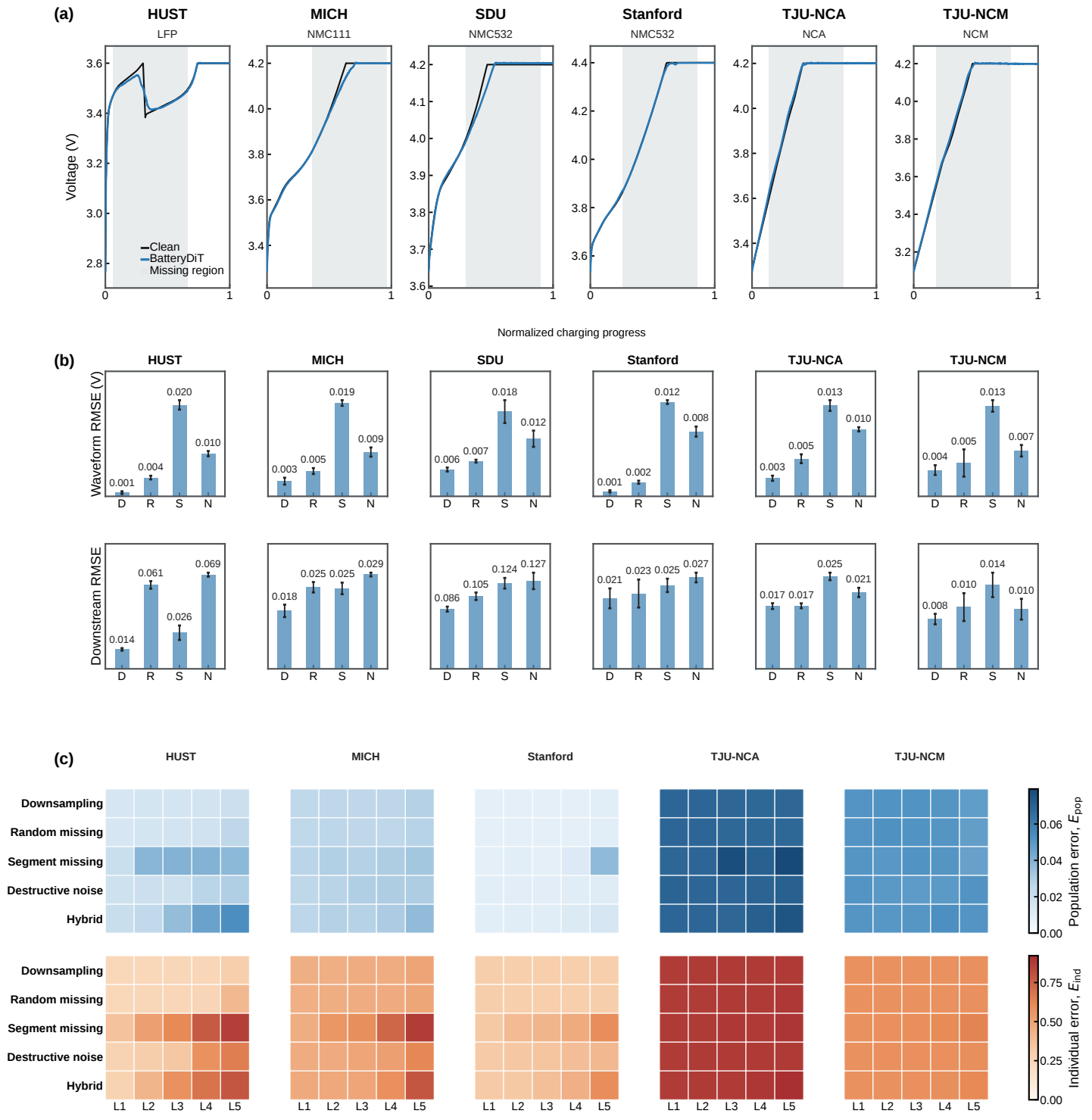

(a)
HUST
LFP
MICH
NMC111
SDU
NMC532
Stanford
NMC532
TJU-NCA
NCA
TJU-NCM
NCM
Voltage (V)
Clean
BatteryDiT
Missing region
Normalized charging progress
(b)
Waveform RMSE (V)
Downstream RMSE
D
R
S
N
(c)
Downsampling
Random missing
Segment missing
Destructive noise
Hybrid
L1
L2
L3
L4
L5
Population error, E_{pop}
Individual error, E_{ind}